THE GREEDY PRIME-RECIPROCAL SUMS OF OEIS A375781: $S_2(42) - S_1(42) = 5$, REFUTING A CONJECTURE OF SIGRIST AND SLOANE

MATH0

Abstract. Let p_k be the k-th prime, let $b(k)$ (OEIS A375781) be the least positive integer keeping $s(k) = \sum_{j \le k} 1/(p_j\, b(j))$ below 1, and write $s(n) = S_1(n)/S_2(n)$ in lowest terms (A375521, A375522). Sigrist and Sloane conjectured that $S_2(n) - S_1(n) = 1$ for all $n \ge 1$ except $n = 4, 6$, having checked $n \le 34$. We show that $S_2(42) - S_1(42) = 5$, so the conjecture is false; to our knowledge this is a new result. As $S_2(42)$ has about $3.9 \cdot 10^{11}$ decimal digits, the proof tracks the numerator of $1 - s(n)$ exactly and its denominator only modulo a shrinking modulus, with a lemma certifying each step. The proof is verified in Lean 4 against a statement that math0 formalized from the OEIS entry, checked against the OEIS data and frozen before the proof attempt.

1. Introduction

Since $\sum_k 1/p_k$ diverges, one can ask how much each term must be damped to keep the partial sums below 1. Rémy Sigrist's sequence A375781 [6] is the lexicographically earliest sequence of positive integers $b(1), b(2), \ldots$ with $s(n) = \sum_{k=1}^{n} 1/(p_k\, b(k)) < 1$ for all $n \ge 1$, where p_k is the k-th prime. It begins $1, 1, 2, 3, 5, 89, 39304, 46994541278, \ldots$ and grows doubly exponentially. Sigrist and Sloane wrote $s(n) = S_1(n)/S_2(n)$ in lowest terms; S_1 and S_2 are A375521 [6] and A375522 [5]. In A375522 they reported that $S_2(n) - S_1(n)$ equals 1 for $1 \le n \le 34$, except at $n = 4$ and $n = 6$, where it equals 2, and conjectured that this pattern continues for all n (Section 2). Equivalently, the remainder $1 - s(n)$ is a unit fraction for every $n \ge 1$ except $n = 4, 6$.

The conjecture is modelled on a theorem about the harmonic series. For the analogous greedy sequence A374663, with k in place of p_k, Sigrist proved that every remainder $1 - s(n)$ with $n > 1$ is the reciprocal of an integer [6, 7]. In A375528 [6], the analogue for the lucky numbers, Sloane writes that for the primes "it seems that all partial sums except for $n = 4$ and 6 have this property". The sequences are discussed in Sloane's talk at the Rutgers Experimental Mathematics Seminar [9]; the slides list them and give no counterexample. They also state a more general conjecture (Conjecture 4 there): for such dampened series there are a constant c and integers e_n with $s(n) = (e_n - c)/e_n$ for all sufficiently large n. Our result does not decide that conjecture. The entries A374663 and A376048 [6] cite a manuscript in preparation by Sigrist and Sloane, *Dampening Down a Divergent Series* [8].

Result and status. We prove that $S_2(42) - S_1(42) = 5$ (Theorem 1), so the conjecture is false. The math0 referees found no earlier counterexample or proof in the revision histories of A375522 and the related entries above, the slides of [9], arXiv (the recent collection [2] of proofs of OEIS conjectures does not treat these sequences), web searches, or the formal-conjectures repository [1], which does not contain this conjecture. The archive of the SeqFan mailing list could not be reached and was not searched. With that caveat, we classify the result as *new*.

The difficulty is size. If $1 - s(n-1) = 1/D$, then $1 - s(n) = (p_n - r)/\big(D(D + p_n - r)\big)$ with $r = D \bmod p_n$ (Remark 6), so the denominators roughly square at each step: $S_2(20)$ has 92406

Date: October 18, 2026.

decimal digits, and $S_2(42)$ has about $3.9\cdot10^{11}$ digits. We never compute $S_2(42)$. Instead we track the small numerator u_n of $1 - s(n)$ exactly and the denominator only modulo a modulus that shrinks at each step (Lemma 7); the Lean kernel runs and checks this computation. Numerical computations outside the formal proof show that $n = 42$ is the smallest counterexample, and that further counterexamples occur at $n = 50$ (difference 3), $n = 56$ and $n = 59$ (difference 5 in both cases), with no others for $n \le 70$.

2. Statement

2.1. **The sequences.** Let p_k denote the k-th prime ($p_1 = 2$). For $x \in \mathbb{Q}$ let $\mathrm{num}(x) \in \mathbb{Z}$ and $\mathrm{den}(x) \in \mathbb{N}_{>0}$ be its numerator and denominator in lowest terms. Define $s(0) = 0$ and, for $n \ge 1$,

$$b(n) = 1 + \left\lfloor \frac{1}{(1 - s(n-1))\, p_n} \right\rfloor, \qquad s(n) = s(n-1) + \frac{1}{p_n\, b(n)}. \tag{1}$$

By Lemma 5 below, $0 < 1 - s(n) \le 1$ for all n, so (1) is well defined. For $c \in \mathbb{N}_{>0}$, $s(n-1) + 1/(p_n c) < 1$ if and only if $c > 1/((1 - s(n-1))p_n)$, i.e. $c \ge b(n)$; as every admissible prefix extends, b is the lexicographically earliest admissible sequence, the OEIS definition of A375781. Put $S_1(n) = \mathrm{num}(s(n))$ and $S_2(n) = \mathrm{den}(s(n))$.

Theorem 1. $S_2(42) - S_1(42) = 5$.

Corollary 2. *The following statement is false: for every $n \ge 1$ with $n \ne 4$ and $n \ne 6$, $S_2(n) - S_1(n) = 1$.*

2.2. **The OEIS conjecture.** The conjecture is a comment in A375522 [5], an entry by Rémy Sigrist and N. J. A. Sloane dated Aug 30 2024 (revision #25, Apr 22 2025). It reads verbatim:

> Let S(n) = Sum_{k = 1..n} 1 / (k*A375781(k)) = S1(n)/S2(n) when reduced to lowest terms, where S1(n) = A375521(n), S2(n) = the present sequence. The differences S2(n) - S1(n) are surprisingly small: for n = 1,2,...,34 the values S2(n) - S1(n) are: 1, 1, 1, 2, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 suggesting the conjecture that they are always 1 except for n = 4 and 6 (compare the Theorem in A374983).

The entry's name and this comment write $1/(k \cdot \mathrm{A375781}(k))$. The data, the worked example $(0/1, 1/2, 5/6, 14/15, \dots)$ and the Maple, Mathematica and Python programs of A375521, A375522 and A375781 all use $1/(p_k \cdot \mathrm{A375781}(k))$. With k read literally the first sum would already be $1/(1 \cdot 1) = 1$, contradicting $S_2(1) = 2$. So "k" is a typo for prime(k), and (1) is exactly the recursion in the entries' Maple and Mathematica programs.

2.3. **The Lean statement.** The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A375522.lean` (namespace `Math0.Statements.A375522`). It was reviewed for fidelity and frozen (SHA-256 `3eec79b4...ced342`) before any proof attempt. The primes are computed by a bounded search (docstrings omitted):

```
def firstPrimeFrom : ℕ → ℕ → ℕ
  | 0, _ => 0
  | fuel + 1, q => if q.Prime then q else firstPrimeFrom fuel (q + 1)
def nextPrime (m : ℕ) : ℕ := firstPrimeFrom (m + 2) (m + 1)
def primeAt : ℕ → ℕ
  | 0 => 2
  | i + 1 => nextPrime (primeAt i)
def prime (k : ℕ) : ℕ := primeAt (k - 1)
```

The sums, the three sequences and the statement are (verbatim):

```
/-- `s n = ∑_{k = 1..n} 1 / (prime k * b k)` with the greedy
`b k = 1 + ⌊1 / ((1 - s (k - 1)) * prime k)⌋` (A375781), `s 0 = 0`. -/
def s : ℕ → ℚ
  | 0 => 0
  | n + 1 => s n + 1 / ((prime (n + 1) : ℚ) *
      ((1 + ⌊1 / ((1 - s n) * (prime (n + 1) : ℚ))⌋ : ℤ) : ℚ))

/-- A375781: `b n = 1 + floor(1/((1 - s(n-1)) * prime(n)))` for `n ≥ 1`. -/
def b (n : ℕ) : ℤ := 1 + ⌊1 / ((1 - s (n - 1)) * (prime n : ℚ))⌋

/-- A375521: `S1 n`, the numerator of `s n` in lowest terms. -/
def S1 (n : ℕ) : ℤ := (s n).num

/-- A375522: `a n = S2 n`, the denominator of `s n = ∑_{k = 1..n} 1 / (prime k * A375781 k)`
in lowest terms. -/
def a (n : ℕ) : ℕ := (s n).den

/-- Conjecture (Sigrist and Sloane, 2024): for every `n ≥ 1` other than `n = 4` and `n = 6`,
`S2(n) - S1(n) = 1`, i.e. `s n = 1 - 1 / a n`. -/
def statement : Prop :=
  ∀ n : ℕ, 1 ≤ n → n ≠ 4 → n ≠ 6 → (a n : ℤ) - S1 n = 1
```

2.4. **How the encoding matches the OEIS.** *Primes.* The file proves `primeAt_eq_nth`: `primeAt i = Nat.nth Nat.Prime i`, using Bertrand's postulate (Mathlib's `Nat.exists_prime_lt_and_le_two_mul`) to guarantee a prime in the search window $m+1,\dots,2m+2$. So `prime k` is Mathlib's k-th prime p_k for $k \geq 1$; `prime 0` is never used. *Sums.* `s` is the recursion (1) over $\mathbb{Q}$ with `Int.floor`; the lemma `s_succ` (proved by `rfl`) states $s(n+1) = s(n) + 1/(p_{n+1}\, b(n+1))$. *Fractions.* A Mathlib rational is stored in lowest terms with a positive denominator, so `S1 n` and `a n` are $S_1(n)$ and $S_2(n)$, with OEIS offset 0. The difference is taken in $\mathbb{Z}$, so truncated subtraction in $\mathbb{N}$ plays no role. *Range.* The comment concerns $n \geq 1$ (at $n = 0$ the difference is 1 anyway). *Data.* The kernel checks, by `decide +kernel`, that the definitions reproduce all 11 listed terms of A375522 (`a_data`), A375521 (`S1_data`) and A375781 (`b_data`), and that the differences for $n = 1,\dots,24$ are $1,1,1,2,1,2,1,1,\dots,1$, as in the comment (`diff_data`). This also confirms the reading of "k" as prime(k).

Corollary 2 is the Lean theorem `Math0.Frontier.A375522.result`, whose type is ¬ `statement`. Theorem 1 is not a separate exported Lean declaration. It follows from the private lemmas `diff_eq`, `T_spec`, `mrun_spec` and `mrun_42` of Section 3, which are kernel-checked in the same file and combined in the proof of `result`.

3. Proof

Throughout, $R_n = 1 - s(n)$, and for integers m and $M > 0$, $m \bmod M \in [0, M)$ denotes the least nonnegative residue.

3.1. Reduction to the remainder.

Lemma 3 (`diff_eq`)**.** *For every n, $S_2(n) - S_1(n) = \operatorname{num}(R_n)$.*

Proof. For $x \in \mathbb{Q}$ we have $\operatorname{den}(1-x) = \operatorname{den}(x)$, since $1 - x = (\operatorname{den} x - \operatorname{num} x)/\operatorname{den} x$ and $\gcd(\operatorname{den} x - \operatorname{num} x, \operatorname{den} x) = \gcd(\operatorname{num} x, \operatorname{den} x) = 1$. Hence $\operatorname{num}(1-x) = (1-x)\operatorname{den}(1-x) = (1-x)\operatorname{den}(x) = \operatorname{den}(x) - \operatorname{num}(x)$. Take $x = s(n)$. □

Lemma 4 (`rat_step`)**.** *Let u, D, p be positive integers and $R = u/D$ (not necessarily in lowest terms). Write $D = upq + r$ with $q = \lfloor D/(up) \rfloor$ and $0 \le r < up$. Then $\lfloor 1/(Rp) \rfloor = q$ and*

$$R - \frac{1}{p\,(1 + \lfloor 1/(Rp) \rfloor)} = \frac{up - r}{Dp(q+1)}, \qquad up - r \ge 1.$$

Consequently, with $g = \gcd\big(up - r,\ Dp(q+1)\big)$, the numerator and denominator of the left-hand side in lowest terms are $(up - r)/g > 0$ and $Dp(q+1)/g$.

Proof. We have $1/(Rp) = D/(up)$, whose floor is q, and $u/D - 1/(p(q+1)) = (up(q+1) - D)/(Dp(q+1))$ with $up(q+1) - D = up - r \ge 1$. The last claim holds because, for positive integers N and M, the fraction N/M in lowest terms is $(N/\gcd(N,M))/(M/\gcd(N,M))$; in Lean this is `Rat.num_mkRat` and `Rat.den_mkRat`. □

Let $\tau_p(u, D)$ denote the pair $\big((up - r)/g,\ Dp(q+1)/g\big)$ of Lemma 4 (Lean: `tstep`). Define $(u_0, D_0) = (1, 1)$ and $(u_n, D_n) = \tau_{p_n}(u_{n-1}, D_{n-1})$ (Lean: `T`).

Lemma 5 (`T_spec`)**.** *For every n, $\mathrm{num}(R_n) = u_n$, $\mathrm{den}(R_n) = D_n$ and $u_n > 0$. In particular $0 < R_n \le 1$, and $S_2(n) - S_1(n) = u_n$.*

Proof. Induction on n. For $n = 0$, $R_0 = 1 = 1/1$. For the step, (1) gives $R_n = R_{n-1} - 1/\big(p_n(1 + \lfloor 1/(R_{n-1}p_n) \rfloor)\big)$, and $R_{n-1} = u_{n-1}/D_{n-1}$ with $u_{n-1}, D_{n-1} > 0$ by the induction hypothesis. Lemma 4 gives $\mathrm{num}(R_n) = u_n > 0$ and $\mathrm{den}(R_n) = D_n$. Moreover $R_n \le R_{n-1} \le 1$. The last assertion follows from Lemma 3. □

Remark 6. Suppose $u_{n-1} = 1$, write $D = D_{n-1}$, $p = p_n$, $r = D \bmod p$, and put $m = p - r$. Then $p(q+1) = D + m$, so $1 - s(n) = m/(D(D+m))$. Since $D(D+m) \equiv D^2 \pmod m$,

$$u_n = \frac{m}{\gcd(m, D^2)}, \qquad D_n = \frac{D(D+m)}{\gcd(m, D^2)}.$$

Hence $u_n = 1$ if and only if $(p_n - r) \mid D_{n-1}^2$, and $D_n \ge D_{n-1}^2/p_n$. The Lean proof does not use this remark.

3.2. **A certified modular simulation.** For a prime p and a triple $(u, d, X) \in \mathbb{N}^3$, where d stands for a residue of the true denominator modulo X, put

$$\nu = up - (d \bmod up), \quad \delta = d\,p\,\big(\lfloor d/(up) \rfloor + 1\big), \quad \gamma = \gcd(\nu, \delta), \quad X' = \big\lfloor \lfloor X/u \rfloor / \gamma \big\rfloor,$$

and define $\mu_p(u, d, X) = \big(\nu/\gamma,\ (\delta/\gamma) \bmod X',\ X'\big)$ (Lean: `mstep`). The step is *checked* if $\mathrm{ok}_p(u, d, X)$ holds, meaning that $up > 0$, $up \mid X$ and $\nu \mid \lfloor X/u \rfloor$ (Lean: `mok`).

Lemma 7 (`mstep_spec`)**.** *Let $p, u, D, d, X \in \mathbb{N}$ with $\mathrm{ok}_p(u, d, X)$ and $D \equiv d \pmod X$. Then the first component of $\tau_p(u, D)$ equals ν/γ, and the second component of $\tau_p(u, D)$ is congruent to δ/γ, hence to $(\delta/\gamma) \bmod X'$, modulo X'.*

Proof. Write $X = upk$. Then $\lfloor X/u \rfloor = pk$, and the hypothesis says $\nu \mid pk$. (a) Since $up \mid X$ and $D \equiv d \pmod X$, we have $D \bmod up = d \bmod up =: r$. So the true unreduced numerator $up - (D \bmod up)$ equals $\nu = up - r$. (b) Write $D = upq_1 + r$ and $d = upq_2 + r$. Then $upk = X$ divides $d - D = up(q_2 - q_1)$, hence $k \mid q_2 - q_1$. Write $d - D = upkc$ and $q_2 - q_1 = ke$ with $c, e \in \mathbb{Z}$. (c) Let $\delta_t = Dp(q_1 + 1)$ be the true unreduced denominator. Then $\delta = dp(q_2 + 1)$, and

$$\delta - \delta_t = p\big[(d - D)(q_2 + 1) + D(q_2 - q_1)\big] = pk\big[upc\,(q_2 + 1) + De\big],$$

so $\delta_t \equiv \delta \pmod{pk}$. (d) Since $\nu \mid pk$, also $\delta_t \equiv \delta \pmod \nu$, so $\gcd(\nu, \delta_t) = \gcd(\nu, \delta) = \gamma$. The true step therefore divides by the same γ, and its first component is ν/γ by Lemma 4. (e) Since γ divides ν, it divides pk, and it also divides δ_t and δ. Cancelling γ in $\delta_t \equiv \delta \pmod{pk}$ gives $\delta_t/\gamma \equiv \delta/\gamma \pmod{pk/\gamma}$, and $pk/\gamma = X'$. □

n	p_n	r	ν	γ	n	p_n	r	ν	γ	n	p_n	r	ν	γ
1	2	1	1	1	15	47	39	8	8	29	109	9	100	100
2	3	2	1	1	16	53	20	33	33	30	113	57	56	56
3	5	1	4	4	17	59	29	30	30	31	127	104	23	23
4	7	1	6	3	18	61	15	46	46	32	131	0	131	131
5	11	17	5	5	19	67	66	1	1	33	137	27	110	110
6	13	11	2	1	20	71	19	52	52	34	139	132	7	7
7	17	33	1	1	21	73	17	56	56	35	149	57	92	92
8	19	17	2	2	22	79	45	34	34	36	151	23	128	128
9	23	15	8	8	23	83	79	4	4	37	157	52	105	105
10	29	1	28	28	24	89	0	89	89	38	163	38	125	125
11	31	28	3	3	25	97	4	93	93	39	167	36	131	131
12	37	11	26	26	26	101	0	101	101	40	173	0	173	173
13	41	21	20	20	27	103	0	103	103	41	179	178	1	1
14	43	22	21	21	28	107	3	104	104	**42**	181	56	125	25

TABLE 1. The checked run from $X_0 = (190!)^2$: $r = D_{n-1} \bmod u_{n-1}p_n$, $\nu = u_{n-1}p_n - r$, $\gamma = \gcd(\nu, \delta)$, $u_n = \nu/\gamma$. Here $u_n = 1$ except at the steps in bold ($u_4 = u_6 = 2$, $u_{42} = 5$). The Lean theorem records only $u_{42} = 5$; the table was recomputed in Python.

Fix $X_0 \in \mathbb{N}$. The *checked run* is $\rho_0 = (1, 1, X_0)$, and $\rho_n = \mu_{p_n}(\rho_{n-1})$ if ρ_{n-1} is defined and $\mathrm{ok}_{p_n}(\rho_{n-1})$ holds; otherwise ρ_n is undefined (Lean: `mrun`, with values in `Option`).

Lemma 8 (`mrun_spec`)**.** *If $\rho_n = (u, d, X)$ is defined, then $u_n = u$ and $D_n \equiv d \pmod X$.*

Proof. Induction on n; for $n = 0$, $(u_0, D_0) = (1, 1)$. If ρ_n is defined, then $\rho_{n-1} = (u, d, X)$ is defined, $\mathrm{ok}_{p_n}(u, d, X)$ holds, and by induction $u_{n-1} = u$ and $D_{n-1} \equiv d \pmod X$. Lemma 7 with $D = D_{n-1}$ gives the claim for $(u_n, D_n) = \tau_{p_n}(u, D_{n-1})$. □

3.3. **The computation.**

Proposition 9 (`mrun_42`)**.** *For $X_0 = (190!)^2$, the checked run ρ_{42} is defined and its first component is 5.*

This is a finite computation carried out by the Lean kernel (Section 4). The modulus X_0 has 2339 bits. Each step divides it by $u_{n-1}\gamma$, and after 42 steps it still has 2157 bits; no intermediate number exceeds about 4700 bits. Table 1 records the run: for each step, the residue $r = D_{n-1} \bmod u_{n-1}p_n$, the unreduced numerator ν, and the cancelled factor γ. By Lemma 7(a),(d) these are the true values, and $u_n = \nu/\gamma$.

Step 42 by hand. The run gives $u_{41} = 1$, and its modulus X_{41} is divisible by $181 \cdot 5^3$; its residue yields $D_{41} \equiv 56 \pmod{181}$ and $D_{41} \equiv 5 \pmod{125}$. So $r = 56$ and $m = \nu = 125$, and by Remark 6, $u_{42} = 125/\gcd(125, D_{41}^2)$. Since $5 \parallel D_{41}$, $\gcd(125, D_{41}^2) = 25$ and $u_{42} = 5$. By contrast, at step 38 the same $\nu = 125$ cancels completely, because $5^2 \mid D_{37}$.

Proof of Theorem 1 and Corollary 2. By Lemma 5, $S_2(42) - S_1(42) = u_{42}$. By Proposition 9 and Lemma 8, $u_{42} = 5$. Since $42 \geq 1$, $42 \neq 4$ and $42 \neq 6$, while $5 \neq 1$, the statement of Corollary 2 fails at $n = 42$. □

The Lean proof of `result` follows this route: it specializes `statement` to $n = 42$, rewrites with `diff_eq` and `T_spec` to get $u_{42} = 1$, and then uses `mrun_42` and `mrun_spec` to get $u_{42} = 5$, a contradiction.

4. Machine verification

Files and declarations. In the math0 repository (`https://github.com/Foundation-0/math0`), the proof is in `lean/Math0/Frontier/OEIS/A375522.lean`, namespace `Math0.Frontier.A375522`, and the main theorem is `Math0.Frontier.A375522.result`. The private auxiliary declarations are named in Section 3 next to the corresponding definitions and lemmas. Verbatim excerpts (docstrings omitted):

```
private def X0 : ℕ := Nat.factorial 190 ^ 2

private theorem mrun_42 : (mrun X0 42).map (·.1) = some 5 := by
  decide +kernel

#math0_settles Math0.Frontier.A375522.result Math0.Statements.A375522.statement
```

Kernel computation. Proposition 9 is proved by `decide +kernel`: the Lean kernel reduces the decision procedure to `true`, computing 190!, the primes $p_1, \dots, p_{42}$ via the statement file's `prime`, and all 42 steps and checks with its built-in arbitrary-precision arithmetic on natural-number literals. No compiled code is trusted; the file uses no `native_decide` (which would add the axiom `Lean.ofReduceBool`), `sorry`, `axiom` or `set_option`.

Audit. The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a universe-monomorphic definition of type `Prop`; (2) `statement` depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`, so it cannot hide a `sorry`; (3) the type of `result` is, up to metadata and with no unfolding, either `statement` or ¬ `statement`; and (4) `result` also depends only on those three axioms. On this file it reports:

```
math0 audit: 'Math0.Statements.A375522.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A375522.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A375522.result' refutes 'Math0.Statements.A375522.statement'
```

Toolchain and reproduction. We use Lean 4 [4] `v4.33.1` and Mathlib [3] `v4.33.1`; the repository also pins formal-conjectures [1] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root; it builds all math0 results with `lake build` (re-running every audit command) and checks that the pinned dependencies are unmodified. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A375522.lean` in `lean/`; this took about one minute of wall-clock time on our machine. Two math0 referees reimplemented the run in Python, with $X_0 = (190!)^2$ and with other starting moduli up to $(2000!)^4$, and obtained identical results.

References

[1] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171, 2026. `https://github.com/google-deepmind/formal-conjectures`.

[2] S. Fried, *Further proofs of conjectures from the OEIS*, arXiv:2607.24832, 2026.

[3] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[5] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A375522 (R. Sigrist and N. J. A. Sloane), revision #25, Apr 22 2025, `https://oeis.org/A375522`.

[6] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entries A375521 (R. Sigrist and N. J. A. Sloane), A374663, A374983 and A375781 (R. Sigrist), A375516, A375528 and A376048 (N. J. A. Sloane), at `https://oeis.org/`.

[7] R. Sigrist, *Proof of Theorem* (in A374983), Aug 26 2024, revised Sep 01 2024, `https://oeis.org/A374983/a374983.txt`.

[8] R. Sigrist and N. J. A. Sloane, *Dampening Down a Divergent Series*, manuscript in preparation, September 2024 (cited in OEIS A374663 and A376048).

[9] N. J. A. Sloane, *A Nasty Surprise in a Sequence and Other OEIS Stories*, Experimental Mathematics Seminar, Rutgers University, Oct 10 2024. Video: `https://www.youtube.com/watch?v=3RAYoaKMckM`; slides: `https://sites.math.rutgers.edu/~zeilberg/expmath/sloane85BD.pdf`.

Produced by Claude (an AI model developed by Anthropic) in the math0 project